

DAZZLER SUPPORT FOR MBASIC

Dazzler commands are in the format:

DZOP “<action>” [, <argument>...]

Where “<action>” is a string defining the action to be taken. Only the first character of the string is significant, but you can use more characters to make it clearer what you are trying to do.

Each action requires zero, one or two arguments as described below.

ACTION	ARGUMENTS	DESCRIPTION
“I” – Initialize	<i>None</i>	Initializes the DAZZLER. This is optional. If omitted, the DAZZLER will be initialized automatically on the first call to any of the other actions.
“P” – Position	X, Y	Sets the cursor to the specified coordinates.
“D” —Dot	<i>None</i>	Displays a dot in the current color at the current cursor position.
“C” —Character	Character code	This calls the CHAR entry point in the graphics library with the specified character code. The character code must be an integer in the range 0 through 127. It can be the ASCII code of a printable character, or a control code as described in the DAZZLER GRAPHICS LIBRARY documentation.
“L” —Line	X, Y	Draws a line in the current cole or from the current cursor position to the specified X, Y coordinate. This will include a dot at each end of the line. The cursor position is moved to X, Y.
“R” —Rectangle	X, Y	Fills in a rectangle in the current color from the current cursor position to the location (X, Y), inclusive. The cursor position and (X,Y) are the two ends of either of the two diagonals of the rectangle, in any order.
“S” —String	String	A string expression of characters to be sent via the CHAR entry point. Each character must have an ASCII code in the range 0 through 127. Each character can be a printable character, or a control code as described in the DAZZLER GRAPHICS LIBRARY documentation.
“Z” —Zap	<i>None</i>	Clears the entire display
“W” —Wait	<i>Time</i>	Waits 0.1 second times the value of “Time”.
“A” --Animate“	<i>None</i>	“Flips” to the other page of display memory. You can write to one page while the other is displayed.
“Y’—cYcle	<i>None</i>	Cycles the display through the palette(??)
DZF	X, Y	Tests for specified color dot at the X,Y position

In addition to the DZOP command, there is a DZF function that returns the current value of a pixel. Its syntax is DZF(X,Y). In color mode, it will return a color code equal to that used to write the pixel, in the range 16 through 31. In monochrome mode it will return a zero or one. There is a special case. If you write with color code 24 (alternate black), it will return 16. This is because the Cromemco supplied graphics library stores black instead of alternate black in the buffer when color code 24 is used.

Note that DZF() always accesses the first page of memory. I could add a command (DZOP “F”,0 or DZOP “F”,1) to select the page used by DZF().

COLOR = DZF(X,Y)

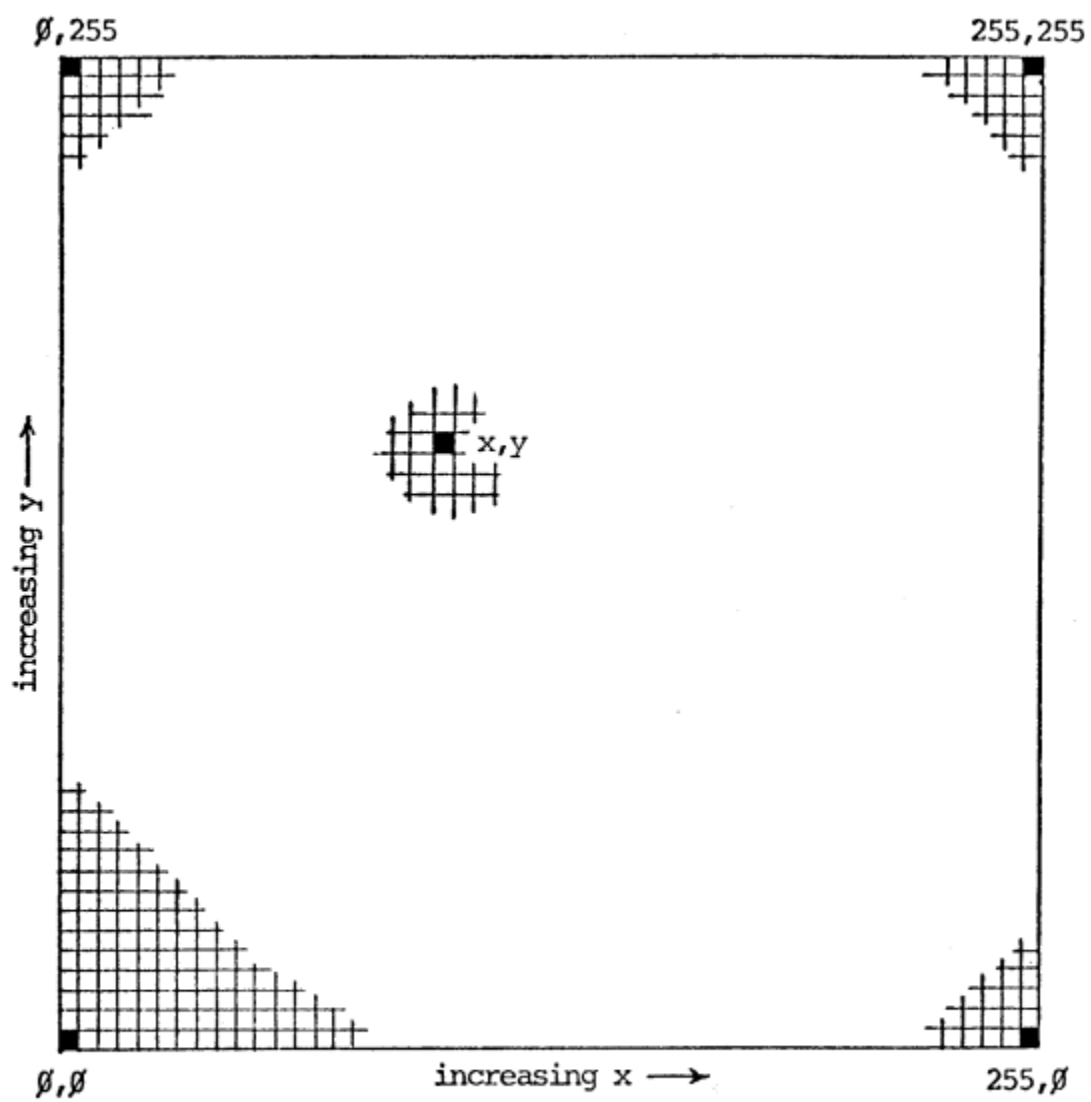


Figure 2-1. The Standard Coordinate System.